

BUSINESS ANALYSIS GUIDE

Amazon CSV Business Questions

A complete guide to answering real business questions using the Amazon Perfect Analytics CSV dataset. Covers sales, customers, products, marketing, logistics, risk, pricing, and seasonal trends with 48 columns and 1 million orders.

Data Analysis Reference

1,000,000 Orders | 48 Columns | 6 Product Categories

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1. Introduction

This guide explains the **Amazon Perfect Analytics CSV file**. The file has **1,000,000 rows** (one million orders) and **48 columns**. It simulates real Amazon e-commerce data from the year 2025. You can use this file to practice data analysis, build dashboards, write SQL queries, and answer real business questions. The data is generated with realistic patterns, so your analysis results will look like real Amazon business data.

The CSV file is created by the Python script **generate_amazon_analytics.py**. This script uses brand-category matching, returning customer models, seasonal trends (Prime Day, Black Friday), regional logistics, risk-based fraud scoring, and category-specific product attributes. Because of these realistic features, the data is very useful for business analysis practice and interview preparation.

1.1 File Overview

Item	Value
File Name	amazon_perfect_analytics_1m.csv
Total Rows	1,000,000
Total Columns	48
Unique Products	10,000
Unique Customers	200,000
Product Categories	6 (Electronics, Computers, Smart Home, Home and Kitchen, Sports and Outdoors, Books)
Date Range	January 2025 - December 2025
File Size	About 500-700 MB

Table 1: File Overview

2. Column Descriptions

The CSV file has 48 columns. Each column has a specific business meaning. Understanding each column is the first step before you start analysis. Below we explain all columns in 8 groups.

2.1 Order Information

These columns tell you basic information about each order. You can use them to count orders, find order dates, and identify unique transactions.

Column	Type	Description
Order_ID	Text	Unique order number (e.g., AMZN-10000001)

Session_ID	Text	Shopping session ID. One session can have many clicks.
Order_Date	Date-Time	When the customer placed the order (YYYY-MM-DD HH:MM:SS)
Actual_Delivery_Date	Date-Time	When the order was delivered. Empty if not delivered yet.
Customer_ID	Text	Unique customer ID (e.g., CUST-100001)
Order_Status	Text	Delivered, Shipped, Processing, Cancelled, or Returned

Table 2: Order Information Columns

2.2 Product Information

Product columns help you understand what was sold. You can analyze which categories are popular, which products have good ratings, and how prices compare across products.

Column	Description
ASIN	Amazon Standard Identification Number. Unique product code.
Product_Category	Electronics, Computers, Smart Home, Home and Kitchen, Sports and Outdoors, Books
Product_Name	Full product name (e.g., Sony Wireless Headphones X-234)
Unit_Price	Price of one unit before discount
COGS_Price	Cost of Goods Sold. How much Amazon paid for the product.
Competitor_Price_At_Order	Price of the same product on competitor websites at order time
Price_Elasticity_Score	How sensitive customers are to price changes (0.5-2.5)
Product_Rating	Customer rating from 3.0 to 5.0 stars
Review_Count	Total number of customer reviews
Buy_Box_Eligible	Yes or No. Can this seller win the Buy Box?

Table 3: Product Information Columns

2.3 Financial Data

Financial columns are very important for business analysis. They help you calculate revenue, profit, margins, and understand how money flows through the business.

Column	Description
Quantity	Number of items ordered (1, 2, 3, 4, or 5)
Discount_Amount	How much was discounted from the original price
Tax_Amount	Tax amount (8% of the base total)
Shipping_Fee	Shipping cost. 0.00 for Prime members, \$4.99-\$7.99 for others
Total_Amount	Final amount the customer paid

Net_Profit	Total Amount minus COGS and shipping. The real profit.
Payment_Method	Credit Card, Amazon Pay, PayPal, Gift Card, Debit Card, Venmo
Promotion_Type	None, Lightning Deal, Coupon, Subscribe and Save, Prime Exclusive, Buy X Get Y

Table 4: Financial Data Columns

2.4 Customer Data

Customer columns help you understand who is buying. You can find Prime vs non-Prime members, which devices they use, and their lifetime value.

Column	Description
Is_Prime_Member	1 = Prime member (free shipping), 0 = not a Prime member
Device_Type	Mobile App (55%), Desktop (30%), Mobile Web (15%)
Customer_State	US state code (e.g., CA, TX, NY) - 30 states
Customer_Region	West, Central, or East
Customer_Lifetime_Value	Total money this customer has spent across all orders
Cart_Abandonment_History	How many times this customer left items in the cart before

Table 5: Customer Data Columns

2.5 Marketing Data

Marketing columns show how customers found the product. You can analyze which ad campaigns are effective and which traffic sources bring the most sales.

Column	Description
Traffic_Source	Google Search, TikTok Ad, Amazon Internal Search, Direct Link, Instagram Influencer, YouTube Review
Keywords_Used	Search keywords the customer typed. Empty for other sources.
Ad_Campaign_ID	Campaign ID if the order came from an ad. Empty if organic.

Table 6: Marketing Data Columns

2.6 Logistics Data

Logistics columns show how orders are shipped and delivered. You can analyze delivery speed, warehouse performance, and shipping carrier effectiveness.

Column	Description
Seller_Type	Sold by Amazon or 3rd-Party Merchant
Warehouse_ID	Fulfillment center code (e.g., FC-DFW-1). 15 warehouses in 3 regions.
Shipping_Carrier	Amazon Logistics, UPS, FedEx, DHL, USPS

Delivery_Days_Estimated	Estimated delivery days. 1-2 for Prime, 3-7 for others.
Lead_Time_Days	Days needed to restock the product (2-14 days)
Package_Weight_kg	Package weight in kilograms (varies by category)
Package_Dimensions_cm	Package size in cm (e.g., 25x15x10)
Hazmat_Status	1 if product contains hazardous materials, 0 if not

Table 7: Logistics Data Columns

2.7 Risk and Returns Data

Risk columns help you identify fraud and predict returns. Fraud scores are based on customer age, payment method, order value, and order history.

Column	Description
Return_Probability_Score	Probability of return (0.0 to 1.0). Higher = more likely to return.
Return_Reason	Why the product was returned: Defective, Wrong Item, Changed Mind, etc.
Fraud_Score	Risk score (0.0 to 1.0). Higher = more risky order.

Table 8: Risk and Returns Columns

2.8 Website Behavior Data

Behavior columns show how customers interacted with the website before buying. Mobile users spend less time, desktop users spend more time researching.

Column	Description
Time_On_Page_Sec	Seconds spent on the product page before ordering
Click_Stream_Count	Number of clicks during the shopping session

Table 9: Website Behavior Columns

3. Sales and Revenue Questions

Sales and revenue analysis is the most common use of this data. Every business needs to know how much money it makes, which products sell well, and how sales change over time. The questions below cover daily, monthly, and category-level analysis.

3.1 Total Revenue

Question: What is the total revenue for the year 2025?

You can answer this by adding all values in the Total_Amount column. This gives you the grand total of all money received from customers. You can also compare this number with the Net_Profit column to see how much profit the business made after product costs.

```
SELECT SUM(Total_Amount) AS Total_Revenue,  
SUM(Net_Profit) AS Total_Profit,  
SUM(Total_Amount) - SUM(Net_Profit) AS Total_Costs  
FROM amazon_perfect_analytics_1m;
```

3.2 Monthly Sales Trend

Question: How do sales change month by month?

This is a very important question. You need to extract the month from Order_Date and group the revenue by month. You will see that November and December have the highest sales because of Black Friday and holiday shopping. July is also high because of Prime Day. January and February are the lowest months.

```
SELECT strftime('%Y-%m', Order_Date) AS Month,  
COUNT(*) AS Orders,  
ROUND(SUM(Total_Amount), 2) AS Revenue,  
ROUND(AVG(Total_Amount), 2) AS Avg_Order_Value  
FROM amazon_perfect_analytics_1m  
GROUP BY Month  
ORDER BY Month;
```

3.3 Revenue by Product Category

Question: Which product categories make the most money?

Grouping by Product_Category shows you which categories drive the most revenue. Computers likely have the highest revenue because their prices are high (\$300-\$1900). But Electronics might have more orders because prices are lower (\$20-\$500). This difference between revenue and order count is important for business strategy.

```
SELECT Product_Category,  
COUNT(*) AS Total_Orders,  
ROUND(SUM(Total_Amount), 2) AS Revenue,  
ROUND(SUM(Net_Profit), 2) AS Profit,  
ROUND(AVG(Total_Amount), 2) AS Avg_Order  
FROM amazon_perfect_analytics_1m  
GROUP BY Product_Category  
ORDER BY Revenue DESC;
```

3.4 Best-Selling Products

Question: What are the top 10 best-selling products by revenue?

Finding the best-selling products helps the business know which products to stock more. You can rank products by total revenue or by order count. Sometimes a product has many orders but low revenue (cheap products). Other products have few orders but high revenue (expensive products). Both views are useful.

```
SELECT Product_Name, Product_Category,  
COUNT(*) AS Orders,  
ROUND(SUM(Total_Amount), 2) AS Revenue,  
ROUND(AVG(Product_Rating), 1) AS Avg_Rating  
FROM amazon_perfect_analytics_1m
```

```
GROUP BY Product_Name, Product_Category
ORDER BY Revenue DESC
LIMIT 10;
```

3.5 Average Order Value

Question: What is the average order value and how does it change by month?

Average Order Value (AOV) is Total_Revenue divided by Number_of_Orders. It tells you how much a typical customer spends per order. If AOV increases over time, it means customers are buying more expensive items or more quantity. You can also compare AOV between Prime and non-Prime members.

3.6 Daily Sales Pattern

Question: Which hours of the day have the most orders?

By extracting the hour from Order_Date, you can see peak shopping hours. You will find that evening hours (18:00-22:00) have the most orders because people shop after work. Night hours (01:00-06:00) have the fewest orders. This information helps schedule ad campaigns and server capacity.

4. Profitability Analysis

Revenue is not the same as profit. A business can have high revenue but low profit if costs are too high. This section shows how to analyze profitability using the financial columns in the CSV file.

4.1 Profit Margin by Category

Question: Which product categories have the highest profit margin?

Profit margin is Net_Profit divided by Total_Amount, shown as a percentage. Books often have high margins (low cost, stable price). Electronics might have lower margins because of competition. Knowing this helps the business decide which categories to focus on for growth.

```
SELECT Product_Category,
ROUND(SUM(Net_Profit), 2) AS Total_Profit,
ROUND(SUM(Total_Amount), 2) AS Total_Revenue,
ROUND(SUM(Net_Profit) * 100.0 / SUM(Total_Amount), 1) AS Profit_Margin_Pct
FROM amazon_perfect_analytics_1m
GROUP BY Product_Category
ORDER BY Profit_Margin_Pct DESC;
```

4.2 Impact of Discounts

Question: How do discounts affect profit?

You can compare orders with discounts vs. orders without discounts. Large discounts increase sales volume but decrease profit per item. The key question is: does the extra sales volume make up for the lower profit? You can group by Promotion_Type to see which promotions are most effective.

4.3 COGS Analysis

Question: What is the total Cost of Goods Sold (COGS) and how does it compare to revenue?

COGS is the cost Amazon pays to buy or make the product. The COGS_Price column shows this for each item. Total COGS = COGS_Price multiplied by Quantity. The difference between Total_Amount and total COGS tells you the gross margin. A healthy business has a gross margin of 30-50%.

5. Customer Behavior Questions

Understanding customers is key to growing any business. The CSV file has customer data that helps you analyze who buys, how often they buy, how much they spend, and which channels they use. There are 200,000 unique customers with repeat purchase behavior.

5.1 Prime vs Non-Prime Members

Question: Do Prime members spend more than non-Prime members?

Prime members get free shipping, which encourages more frequent purchases. You can compare total spending, average order value, and order frequency between Prime (Is_Prime_Member = 1) and non-Prime (Is_Prime_Member = 0) customers. Prime members typically spend 2-3 times more per year than non-Prime members.

```
SELECT Is_Prime_Member,
COUNT(DISTINCT Customer_ID) AS Customers,
COUNT(*) AS Orders,
ROUND(AVG(Total_Amount), 2) AS Avg_Order,
ROUND(SUM(Total_Amount), 2) AS Total_Revenue
FROM amazon_perfect_analytics_1m
GROUP BY Is_Prime_Member;
```

5.2 Top Customers by Lifetime Value

Question: Who are the top 20 most valuable customers?

Customer_Lifetime_Value (CLV) shows the total amount a customer has spent. Finding the top customers helps with loyalty programs and personalized marketing. These customers are the most important to keep happy because they generate the most revenue.

```
SELECT Customer_ID, COUNT(*) AS Orders,
ROUND(SUM(Total_Amount), 2) AS Total_Spent,
Customer_Lifetime_Value AS CLV,
Is_Prime_Member, Customer_State
FROM amazon_perfect_analytics_1m
GROUP BY Customer_ID, Customer_Lifetime_Value, Is_Prime_Member, Customer_State
ORDER BY Total_Spent DESC
LIMIT 20;
```

5.3 Repeat Purchase Rate

Question: What percentage of customers made more than one purchase?

A repeat purchase rate shows customer loyalty. First, count total orders per customer. Then count how many customers have 2 or more orders. Divide by total customers. A good repeat rate for e-commerce is 20-40%. In our data, the returning customer model ensures realistic repeat rates based on customer activity levels.

5.4 Customer Demographics by Region

Question: Which US regions generate the most sales?

The CSV has Customer_State and Customer_Region columns. Grouping by region (West, Central, East) shows where customers live. California (West) likely has the most orders because it is the most populous state. But per-capita spending might be higher in other states.

5.5 Device Preference

Question: Do customers on mobile buy differently from desktop users?

Device_Type shows whether the customer used Mobile App, Desktop, or Mobile Web. Mobile App users buy faster (less time on page) and prefer cheaper items. Desktop users spend more time researching and buy more expensive items. This insight is important for UX design and marketing strategy.

6. Product Performance Questions

Product analysis helps the business decide which products to promote, which to discount, and which to remove from the catalog. The CSV file has 10,000 unique products across 6 categories.

6.1 Product Rating vs Sales

Question: Do higher-rated products sell more?

You can group products by rating ranges (3.0-3.5, 3.5-4.0, 4.0-4.5, 4.5-5.0) and compare order counts. Higher-rated products should have more orders because the generator uses popularity weights based on ratings. This is a common business question in real e-commerce.

6.2 Price Sensitivity

Question: How does price affect order quantity?

In the data, expensive products (over \$500) usually have quantity = 1. Cheaper products (under \$100) often have quantity = 2 or 3. You can verify this by analyzing the relationship between Unit_Price and Quantity. Price_Elasticity_Score also shows how sensitive each category is to price changes.

6.3 Competitor Price Comparison

Question: Are our prices competitive compared to competitors?

Compare Unit_Price with Competitor_Price_At_Order for each category. If our prices are consistently higher, we may lose customers. If our prices are lower, we may win more Buy Box placements but earn less profit. The ideal strategy depends on the product category and market position.

6.4 Buy Box Performance

Question: What percentage of orders have Buy Box eligibility?

The Buy_Box_Eligible column shows Yes or No. Winning the Buy Box is critical on Amazon because most customers click the Buy Box button. Analyze Buy Box eligibility by category and seller type to understand competitive positioning.

7. Marketing and Traffic Analysis

Marketing analysis tells you where your customers come from and which channels are most effective. The CSV has traffic source, keywords, and campaign data.

7.1 Traffic Source Effectiveness

Question: Which traffic source brings the most revenue?

Group by Traffic_Source and compare revenue, order count, and average order value. Amazon Internal Search likely brings the most orders because customers already know what they want. TikTok Ads and Instagram Influencers may bring fewer orders but at higher customer acquisition cost.

```
SELECT Traffic_Source,  
COUNT(*) AS Orders,  
ROUND(SUM(Total_Amount), 2) AS Revenue,  
ROUND(AVG(Total_Amount), 2) AS Avg_Order  
FROM amazon_perfect_analytics_1m  
GROUP BY Traffic_Source  
ORDER BY Revenue DESC;
```

7.2 Keyword Analysis

Question: What are the most common search keywords?

The Keywords_Used column contains search terms. Analyzing the most frequent keywords helps with SEO strategy and ad targeting. You can also see which keywords lead to higher-value orders.

7.3 Campaign ROI

Question: Which ad campaigns are most profitable?

Orders with an Ad_Campaign_ID come from paid ads. Compare the revenue and profit from these orders against the total to measure campaign effectiveness. High revenue from TikTok Ads means the TikTok marketing strategy is working well.

7.4 Promotion Effectiveness

Question: Which promotion types drive the most sales?

Group by Promotion_Type to see which deals are most popular. Lightning Deals and Coupons may drive high volume but lower profit per item. Subscribe and Save brings steady, predictable revenue. Prime Exclusive Discounts encourage Prime membership sign-ups.

8. Logistics and Delivery Questions

Fast and reliable delivery is very important for customer satisfaction. The CSV has detailed logistics data including warehouses, carriers, and delivery times.

8.1 Delivery Speed Analysis

Question: What is the average delivery time by region and carrier?

Calculate the difference between Actual_Delivery_Date and Order_Date for delivered orders. Compare across regions (West, Central, East) and carriers (Amazon Logistics, UPS, FedEx, etc.). Same-region orders should be faster. Prime members get 1-2 day delivery.

8.2 Late Delivery Rate

Question: What percentage of deliveries are late?

A delivery is late if Actual_Delivery_Date is after the estimated date. The data has a 15% late delivery rate, which is realistic for e-commerce. High late rates in specific regions or carriers indicate logistics problems.

8.3 Warehouse Performance

Question: Which warehouses handle the most orders and have the fastest delivery?

There are 15 warehouses across 3 regions. Analyze order volume and delivery speed per warehouse. Some warehouses may be overloaded while others have extra capacity. This helps with warehouse expansion and load balancing decisions.

8.4 Shipping Cost Analysis

Question: How much does the company spend on shipping?

Prime members get free shipping, but the company still pays the carrier. Non-Prime members pay \$4.99-\$7.99. Calculate total shipping costs by carrier and compare with shipping revenue from non-Prime members.

9. Risk and Fraud Analysis

Risk analysis is critical for e-commerce. Fraud costs businesses billions of dollars every year. The CSV has fraud scores based on realistic risk factors like customer age, payment method, order value, and order history.

9.1 Fraud Detection

Question: How many orders have high fraud risk?

Orders with Fraud_Score higher than 0.5 are high risk. The score is based on multiple factors: new customers (joined in 2025), low lifetime value, high order amount, and Gift Card payments. You can filter by these factors to understand the fraud patterns.

```
SELECT COUNT(*) AS High_Risk_Orders,  
ROUND(SUM(Total_Amount), 2) AS At_Risk_Revenue,  
AVG(Fraud_Score) AS Avg_Fraud_Score  
FROM amazon_perfect_analytics_1m  
WHERE Fraud_Score > 0.5;
```

9.2 Return Analysis

Question: Which products and categories have the highest return rates?

The Return_Probability_Score predicts how likely a product is to be returned. Returned orders have a Return_Reason column with specific reasons. Electronics have higher return rates than Books. Common reasons include Defective, Wrong Item, and Changed Mind.

9.3 Cancellation Analysis

Question: What is the cancellation rate and why do customers cancel?

Cancelled orders have higher fraud scores on average. Analyze cancellation patterns by category, payment method, and customer type. High cancellation rates may indicate pricing problems, stock issues, or fraud attempts.

10. Seasonality and Trends

Seasonal trends are very important for inventory planning, staffing, and marketing budgets. The CSV data includes realistic seasonal patterns for the entire year 2025.

10.1 Peak Shopping Days

Question: When are the biggest shopping days of the year?

The data simulates major shopping events. Prime Day (July 15-16) has 3-5x normal sales. Black Friday week (late November) has the highest sales of the year. December 1-15 has high holiday shopping, but after December 15, orders drop because packages cannot arrive before Christmas.

10.2 Day of Week Pattern

Question: Do people buy more on weekends or weekdays?

Extract the day of week from Order_Date and count orders per day. Weekend days (Saturday and Sunday) typically have 20-30% more orders than weekdays. This is because people have more free time to browse and shop on weekends.

10.3 Hourly Pattern

Question: What are the peak shopping hours?

Extract the hour from Order_Date. Peak hours are 18:00-22:00 (after work). Lowest hours are 02:00-07:00 (sleeping time). This pattern helps schedule marketing emails, ad campaigns, and server maintenance.

10.4 Seasonal Inventory Planning

Question: How should inventory be planned for each season?

Combine monthly sales data with Lead_Time_Days to plan inventory. Before peak months (November, December, July), increase stock by 2-3x normal levels. Use the Delivery_Days_Estimated and Warehouse_ID to plan which warehouses need more stock.

11. Pricing Strategy Questions

Pricing is one of the most important business decisions. The right price maximizes both revenue and profit. The CSV has multiple pricing columns for deep analysis.

11.1 Price Distribution by Category

Question: What is the price range for each product category?

Each category has a realistic price range. Books are \$8-\$46, Computers are \$30-\$1900. Understanding the price distribution helps with competitive positioning. You can find the minimum, maximum, average, and median price per category.

11.2 Discount Impact on Sales Volume

Question: Do discounts increase total sales enough to offset lower profit?

Compare orders with 0% discount vs. orders with 10%, 20%, 30% discount. Higher discounts should increase order count but decrease per-order profit. The break-even point is where extra volume makes up for lower margins.

11.3 Price Elasticity Analysis

Question: Which categories are most sensitive to price changes?

Price_Elasticity_Score shows how much demand changes when price changes. A score above 1.5 means demand drops quickly when price increases. A score below 1.0 means demand is stable even with price changes. Use this to decide discount strategies per category.

12. SQL Practice Ideas

This CSV file is excellent for SQL practice. Below are practice ideas organized by difficulty level. Start with beginner questions and move to advanced ones.

12.1 Beginner Level (Basic SELECT)

- Find all orders from California (Customer_State = "CA").
- List all orders where Total_Amount is greater than \$500.
- Show Product_Name and Total_Amount for all Delivered orders.
- Count how many orders were cancelled.
- Find the average Product_Rating for Electronics category.
- List all unique Traffic_Source values.

12.2 Intermediate Level (GROUP BY, JOIN concepts)

- Find monthly revenue for each product category.
- Compare average order value between Prime and non-Prime members.
- Find the top 5 customers by total spending.
- Calculate profit margin percentage for each Promotion_Type.
- Count orders per Traffic_Source and sort by revenue.
- Find the average delivery time by Shipping_Carrier.

12.3 Advanced Level (Subqueries, Window Functions)

- Find products where the actual price is higher than the competitor price.
- Rank customers by lifetime value within each US region.
- Calculate month-over-month revenue growth rate.
- Find customers whose order frequency increased from Q1 to Q4.
- Identify orders where fraud score is in the top 1% of all orders.
- Create a running total of daily revenue.

13. Dashboard and Visualization Ideas

If you want to build a dashboard using tools like Tableau, Power BI, or Python matplotlib, here are the best charts to create with this data.

Chart Type	What to Show	Columns to Use
Line Chart	Monthly revenue trend	Order_Date, Total_Amount
Bar Chart	Revenue by category	Product_Category, Total_Amount
Pie Chart	Traffic source share	Traffic_Source (count)
Scatter Plot	Rating vs. order count	Product_Rating, Quantity
Heatmap	Orders by hour and day	Order_Date (hour, weekday)
Stacked Bar	Status by category	Order_Status, Product_Category
Box Plot	Price range per category	Unit_Price, Product_Category
Histogram	Fraud score distribution	Fraud_Score
Funnel Chart	Cart to purchase rate	Cart_Abandonment, Order_Status
Geo Map	Sales by US state	Customer_State, Total_Amount
Gauge Chart	On-time delivery rate	Actual_Delivery vs. Estimated

Table 10: Dashboard Visualization Ideas

14. Summary

This CSV file is a powerful tool for business analysis practice. It contains 1 million realistic orders with 48 columns covering sales, customers, products, marketing, logistics, risk, and pricing data. You can use it for SQL practice, dashboard building, data analysis projects, and interview preparation.

The key business areas you can explore with this data include: total revenue and profit analysis, monthly and seasonal trends, customer segmentation (Prime vs. non-Prime, device type, geographic region), product performance comparison, marketing channel effectiveness, delivery speed and warehouse optimization, fraud detection and return analysis, and pricing strategy optimization.

Start with simple questions like total revenue and best-selling products. Then move to complex analyses like customer lifetime value prediction, seasonal inventory planning, and fraud pattern detection. Each question in this guide can be answered with SQL, Python, or any BI tool.